

Illumi-Naughty? The Effect of Nighttime Luminosity on Crime in Vancouver

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I. Introduction

Premeditated urban crime remains a central political and policy concern in Vancouver, with an overall 12% increase in the violent crime rate in recent years, despite overall incident counts falling in 2024 (Howell, 2024; Vancouver Police Department, 2024). At the same time, a cooling municipal economy is leading to tightening of municipal budgets, while residents demand safer streets, resulting in a need for cost-effective crime prevention tools. Traditional levers such as increased policing, stricter laws, and greater surveillance, are either expensive or slow to deploy, resulting in a need for more creative nudges. Crime-prevention-through-environmental-design (CPTED) research shows that small tweaks in environmental factors, such as increased lighting, and changes in infrastructure can serve as powerful nudges towards shifting the expected utilities of perpetrators considering committing crimes by changing both the probabilities of detection and the potential payoffs for offending (Chalfin et al., 2021; Cozens et al., 2005).

Building upon this literature, our paper aims to examine if additional nighttime luminosity is an effective measure in deterring premeditated property crime in the city of Vancouver. Unlike the previous literature, which looks either exclusively at the effects of street lighting or changes in evening/morning luminosity induced by daylight savings time (DST) adjustments, satellite radiance captures luminosity values across every block in the city across all hours of the night, allowing the proper capturing of luminosity spillovers induced by street lighting, as well as brightness surrounding commercial activity, seasonal events, and changes in more generalized infrastructure. Our contribution is fourfold:

- 1) We merge monthly NASA Black Marble satellite data for the city of Vancouver with Vancouver Police Department (VPD) naturalized intentional crime data to construct a census-tract-by-month panel for 2014-2023.
- 2) By covering the entire Vancouver city area rather than a handful of treatment blocks, we provide an analysis that is directly relevant to municipal crime prevention efforts.

- 3) We exploit month-to-month variation in nighttime luminosity and employ a fixed-effects negative-binomial model that controls for daytime luminosity as a proxy for economic activity, additional economic covariates, and two-way fixed effects, mitigating omitted-variable concerns.
- 4) By interacting luminosity with a “Downtown” indicator variable, and re-estimating models by offence category, we reveal where and for which crimes nighttime lighting delivers the greatest benefits in reducing the crime rate.

Overall, we find that a 10% increase in nighttime luminosity results in a 2% ($p < 0.001$) decrease in crime, with more pronounced effects for breaking-and-entry crimes, which decrease by 3% ($p < 0.001$). Looking at non-downtown areas, we find that a 10% increase in nighttime luminosity results in a 2.7% decrease in crime ($p < 0.001$).

A. Literature review

Numerous studies have attempted to draw causal links between luminosity and crime rates. For instance, Chalfin et al (2021) study the effects of nighttime luminosity on crime rates in New York City public housing developments, in collaboration with the New York City Housing Authority. The authors conducted a natural experiment by randomly installing 397 temporary lighting towers (TLTs) across 40 housing developments (with an additional 40 housing developments serving as controls) across NYC, which emitted approximately 600,000 lumens (compared to the 5,000 to 35,000 lumens experienced by housing developments without TLTs) and were automatically set to illuminate from dusk until dawn. The authors employed a DDD-style model with 40 randomly selected treatment communities among 340 housing developments, with the non-treated communities representing the control group. Each of those 40 communities received a baseline number of two lighting towers, with additional lighting towers allocated randomly, creating random variation in light counts for treated housing developments.

Furthermore, since crime events are non-negative counts, and typically rare and highly dispersed, the authors implement a Poisson regression model on the crime rate, modelling the logarithm of the expected count as a linear combination of covariates in order to quantify how changes in the dosage of lighting create percent-changes in light counts. The authors found a 35% decrease in the number of nighttime outdoor crimes (90% CI [8–55]), as well as a 5% decrease in serious crime within the treated communities. Which provides significant experimental evidence that Crime Prevention Through Environmental Design (CPTED)-based street lighting treatments may provide significant and inexpensive reductions in crime. However, the experiment-based analysis model is limited by its small sample size, with only 40 treated sites. Hence, the outcomes are highly variable, creating challenges for the precise estimation of the treatment effects. The authors attempt to overcome this by re-randomization and careful covariate selection

using LASSO, but this does not fix underlying heterogeneity issues that may be present; namely, the authors argue for internal validity by controlling for possible heterogeneity via using subsidized housing blocks with the same layout, as well as control for heterogeneity in lighting dosage by employing an additional dosage model. However, the results are still sensitive to the specific model selection, and the authors note that spatial spillover heterogeneity is not likely to be properly controlled for, noting limited statistical power in their heterogeneity controls.

Conversely, Doleac and Sanders (2015) study the effects of luminosity on robbery crime rates using National Incident-Based Reporting System (NIBRS) data, and switches to and from daylight saving time across the United States, which introduces an exogenous variation in luminosity via a sudden shift in the timing of daylight. This discontinuity allows for the implementation of regression discontinuity (RD) models, with time as the running variable, and DST as the discontinuity. To combat heterogeneity introduced by the fact that DST always begins on a Sunday and may coincide with other calendar-specific events, the authors introduce a secondary DiD model that relies on one-hour DST extensions that occurred in 2007 and 2008. The authors find that DST results in a 27% decrease in robbery crimes following the introduction of daylight savings. However, as Domínguez & Asahi (2019) note, the results lack external validity, as most of the districts covered by the NIBRS data are rural, low-population localities.

Domínguez & Asahi (2019), on the other hand, focus specifically on urban areas, and, in similar fashion to Doleac and Sanders (2015), use exogenous variations in luminosity during sunrise and sunset due to daylight savings time (DST) in Santiago, Chile to analyze the effects of luminosity on crime rates, via a regression discontinuity analysis, with two discontinuities during march and october DST changes. The authors employ a regression discontinuity model alongside a DiD complementary analysis, which aims to account for the fact that crime incidence rates may be affected by the time of year itself, since seasonality is inherently wrapped around the time surrounding DST changes, creating multicollinearity. The DiD analysis hence allows for a cleaner analysis of whether crime rates vary simply because of a given season, or if changes in crime rates are truly due to DST-induced changes in luminosity.

The authors find an overall 20% increase in crime rates due to the decrease in luminosity during evening hours during the fall DST transition, as well as a 20% decrease in crime rates during the spring DST transition, noting significant increases in robbery rates during the fall DST transition. Importantly, the authors do not establish an exact causal mechanism behind the changes in crime rates, as they do not identify the behavioral changes causing perpetrators and victims to act differently during DST periods. The authors do test heterogeneity via commuting patterns and land use activity, but other potential sources of het-

erogeneity remain, such as police patrol patterns and psychological behavior.

To our knowledge, the published papers studying luminosity’s effect on crime, as well as the papers studying CPTED more generally, argue that the crime rate is a combination of supply and demand: the likelihood that a criminal commits a crime is affected by the expected severity of the punishment if caught, the potential “gains” a criminal might receive from committing a given crime, as well as the expected likelihood that a criminal is to be caught. Luminosity measures, and CPTED measures more broadly, aim to reduce the latter. Increased luminosity, for instance, creates increased visibility and consequently an increased likelihood for criminals to be spotted by law enforcement, bystanders, and surveillance tools. Likewise, increased luminosity affects the behavior of victims (the “demand-side” of the equation). More luminosity means individuals are more confident in going outside, as well as less weary of their surroundings, providing more opportunities to be victims of an intentional crime.

However, while the aforementioned papers innovate by utilizing natural experiments and exploiting exogenous temporal shifts, they are often constrained by either limited sample sizes (such as in Chalfin et al. (2021)) or exclusively focus on luminosity changes confined to sunrise/sunset periods and not pure nighttime. To address these limitations, our study leverages comprehensive data from NASA’s Black Marble luminosity imaging data satellite, combined with crime statistics from the Vancouver Police Department (VPD). This approach enables us to assess the relationship between pure nighttime luminosity and criminal activity with a larger temporal sample size. Specifically, we hypothesize that increased nighttime luminosity in Vancouver reduces the crime rate during nighttime hours, which we aim to show via robust causal inference methods. Ultimately, we aim to demonstrate that increased luminosity may serve as an effective, affordable deterrent against intentional crime.

Building upon these considerations from previous research, our study takes a similar approach in employing statistical pre-processing suited specifically for crime data. As shown previously, crime data often involve infrequent counts that can appropriately be modeled using either Poisson or negative binomial regression. However, the Poisson regression model assumes equidispersion, a condition rarely met by real-world criminological data. As the previous studies highlight that the crime data commonly exhibit overdispersion, violating this fundamental assumption (Osgood, 2000; MacDonald & Lattimore, 2010), negative binomial regression which explicitly accounts for this excess variability, is often preferred. Given these literature reviews, we considered both Poisson regression and negative binomial regression as our pre-processing methodologies. Detailed methodology selection and comparison between Poisson regression and negative binomial regression will be further discussed in the methodology section.

In addition to our pre-processing methodology choices, we have conducted a careful selection of covariates to be included in our regression. Following the methodological approach of Doleac and Sanders (2015), we include daytime crime rates as a comparative baseline for nighttime crime. Daytime crime serves as a natural control, capturing general crime trends to account for time-varying confounders such as policing intensity and seasonal effects. Comparing changes in crime between daytime and nighttime allows us to more precisely isolate the influence of nighttime luminosity on criminal behavior. Furthermore, literature indicates that daytime crime levels can serve as a proxy for aggregate daily movement patterns. Felson & Boivin (2015) found that areas that experience heavy inflows of individuals such as commuters or shoppers see higher rates of both property and violent crime. Likewise, Paramasivan et al. (2024) found a significant positive correlation between mobility and violent crimes. However, evidence for a direct causal link remains scant.

The choice of covariates in our analysis is further guided by practical considerations related to data availability and fundamental reasoning. For instance, as censuses are typically being conducted at five- or ten-year intervals typically, instead of incorporating panel census data we considered cross-sectional values of demographic and socioeconomic census data as covariates. The United Nations Department of Economic and Social Affairs Statistics Division also notes in the Handbook on the Management of Population and Housing Censuses that frequent censuses are resource-intensive and may not yield significantly different population and housing data from year to year, justifying reliance on the most recently available census. Our panel regression analysis adheres to this practice, incorporating the two most recent cross-sectional census data to control for demographic and socioeconomic characteristics. We next elaborate on each covariate and the rationale for its inclusion. The basic distributional characteristics and summary statistics describing the demographic and socioeconomic covariates appear in Appendix, while the Methodology and Analysis section focuses on how nighttime luminosity relates to crime rates once these covariates are taken into account.

In order to account for the significant influence of urban geographic characteristics on crime patterns, we include a binary covariate indicating whether a neighborhood is located in the downtown area. Sarah L. Boggs (1965), in her study “Urban Crime Patterns,” illustrates that crime distribution significantly varies across city zones, with central business districts exhibiting distinct crime patterns compared to predominantly residential areas. Given that urban location significantly shapes opportunities for criminal activity as shown in the research, we also consider a covariate to differentiate crime patterns unique to downtown Vancouver from those occurring elsewhere in the city.

Lastly, population density was also considered as another key covariate due to its structural influence on community dynamics and crime rates. Battin and Crowl (2017) highlight that population density is intricately linked with crime, as it affects the strength of neighborhood social institutions and informal social controls. The study argues that population density exhibits a negative relationship with property crime, challenging conventional assumptions that denser populations inherently produce higher crime rates. Including population density as a control variable therefore allows us to adjust for these structural effects ensuring that observed variations in crime rates are not attributed to underlying demographic factors.

II. Methodology

A. Data Processing

To better investigate the effect of luminosity on nighttime crime, our study examined two spatial scales: 989 census divisions and 22 neighborhoods delineated from the local area data provided by the City of Vancouver. For both scales, we employed comparable regression specifications. This dual approach facilitates a comprehensive assessment of both broad spatial trends and localized variations in the relationship between luminosity and crime.

Since our study focuses on nighttime crime, it is critical to have an accurate delineation of day and nighttime crime. We used detailed time information from the VPD Open Data alongside historical sunrise and sunset data provided by the `sunalc` R package for the day/night division of crimes. By excluding crimes classified as Mischief and Car Collisions, we focused on intentional offenses. We then matched crime incidents with their corresponding geospatial units using geological data and computed monthly nighttime and daytime crime counts, respectively. With covariate data available at two time points (2016 and 2021 census), we obtained the panel data of two partial scales that span 10 years, from 2014 to 2023. By filling in zeros for months with no crime, the complete data covers all census subdivisions and neighborhoods.

To capture exogenous variation in ambient light, we retrieved NASA's Black Marble VNP 46A3 monthly composite - a 15-arc-second (~ 500 m) grid that is atmospherically corrected, cloud-, lunar-, and BRDF-adjusted, providing markedly cleaner radiance than earlier series. Using the `blackmarbler` R package, we downloaded for 2012-2023, mosaicked the LAADS-DAAC tiles, and aggregated the Near-Nadir Composite Snow-Free night-time light over our study polygons. We fetched monthly mean NTL within 989 census-division boundaries and, for greater robustness, within 22 neighborhood polygons; the latter span dozens of VI-

IRS pixels, whereas several census-division units align with only a single pixel and are thus more susceptible to residual noise. The Black Marble algorithm’s rigorous lunar-irradiance modelling, stray-light removal, and temporal gap-filling further enhance radiometric stability and temporal consistency, validated in NASA’s official Algorithm Theoretical Basis Document (Román et al., 2018).

B. Modeling Decisions and Assumption Checks

In the first stage of our research, we tried to implement a DID model with continuous treatment on our data; however, we faced three major challenges. First, there was no significant policy breaking point for us to identify a treatment period, as although we observed an overall slow upward trend in the monthly average NTL from both scales, there was no obvious similarity in trends to support a parallel trend assumption and identify a treatment and control group.

Second, the distribution of observed NTL is highly volatile conditional on months and census divisions/neighborhoods, which puts a question about whether or not they were truly following an increasing trend individually. Third, although we detected some structural breaks within the data, there was not enough policy evidence for us to believe the treatment was intentional.

Therefore, we argue that the allocation of NTL should be random conditional on time, region, and a set of additional covariates. Consequently, we switched to a panel fixed effect model, which better aligns with our panel data featuring a continuous treatment variable. This framework also partially mitigates concerns regarding Omitted Variable Bias (OVB), endogenous bias, and reverse causality. Nonetheless, the decision subjects us to the potential selection bias introduced by the non-random assignment of treatment at the given time t and region i , which requires us to investigate the bias carefully.

In our empirical strategy, we addressed the incomparability bias B_1 by excluding observations with missing values in our data. After dealing with missing values, we have a total of 35 divisions out of 989 at the census division level that do not have covariate data for each month of the full decade, while the data for neighborhoods are fully balanced as shown in the Appendix Table A3. This also provides a basis for our conclusion in favor of neighborhood level. Furthermore, we assessed the imbalance bias B_2 and selection bias B_3 by creating the balancing table (Table 1) for the covariates, stratified by the “high” and “low” values separated by the median of the treatment variable. The observation of statistically significant differences in the means of certain covariates affirmed our decisions to include covariates. Additionally, to capture the potential heterogeneous effects introduced by a different crime pattern in the downtown area, we included the indicator of “Downtown” to control for the regional effect. We also decided to use daytime crime as a control to better isolate changes in nighttime crime caused by

Table 1—: Covariate Balance by Treatment Status

Variable	High	Low	p-value	SMD
Census Division Covariates				
Downtown	0.10 (0.30)	0.00 (0.03)	<0.001	0.465
Electricity Price Index	115.39 (7.27)	114.81 (7.17)	<0.001	0.080
Average Income (scaled by \$100,000)	0.73 (0.23)	0.90 (0.27)	<0.001	0.654
Expected Day Crime	1.41 (4.62)	0.40 (0.70)	<0.001	0.307
Population Density	13,567.03 (11,995.65)	6,638.29 (5,007.72)	<0.001	0.754
Immigrant Proportion	0.38 (0.14)	0.45 (0.14)	<0.001	0.471
University Proportion	0.18 (0.20)	0.18 (0.17)	0.002	0.019
School Index	2.75 (0.75)	2.49 (0.85)	<0.001	0.333
Bus Index	3.14 (0.73)	2.68 (0.77)	<0.001	0.620
Cultural Space Index	2.63 (0.98)	1.79 (0.67)	<0.001	1.007
Mean Land Value (scaled by \$1 million)	1.81 (3.81)	2.10 (3.87)	<0.001	0.076
Neighborhood Covariates				
Downtown	0.09 (0.29)	0.00 (0.00)	<0.001	0.447
Average Income (scaled by \$100,000)	0.74 (0.17)	0.95 (0.19)	<0.001	1.136
Daytime Crime Count	61.00 (74.58)	17.98 (13.30)	<0.001	0.803
Population Density	8,619.08 (5,967.44)	4,300.90 (1,926.55)	<0.001	0.974

Notes: Means (standard deviations in parentheses) by treatment status; p-values from two-sided t-tests; standardized mean differences (SMD). High = above median; Low = below median. Electricity Price Index derived from EPSPI (Statistics Canada); other indices based on a linear proximity algorithm on Canadian Census data. The final four variables are calculated at the neighborhood level.

other factors. Together, these methodological decisions bolster our assumption of conditional independence.

A key assumption needed to further affirm the conditional independence assumption is that the “Downtown” covariate and daytime crime patterns serve as an effective proxy for both economic activity and mobility patterns. The latter is a covariate widely used in the literature surrounding luminosity and crime, as both Domínguez & Asahi (2019) and Doleac and Sanders (2015) use economic controls in their regression-based analyses. The literature review extensively covers the existing literature surrounding using daytime crime as a proxy for mobility patterns. Our analysis hence operates under the assumption that daytime crime causes changes in mobility patterns, which in turn cause changes in economic activity. The addition of the ‘is-downtown’ covariate serves as an additional control for downstream mobility and economic activity patterns. However, as we will discuss in our extension, this assumption may not entirely hold. Regardless, given the importance of including economic controls, we argue that the inclusion of two imprecise controls is better than omitting them entirely.

C. Models of Two Scales

CENSUS DIVISIONS

Given the prevalence zero-inflated crime counts and the over-dispersion in monthly crime data on the census division level, we employed negative binomial regressions (NB) for nighttime and daytime offences, incorporating year-

Table 2—: AIC and BIC Comparison of Potential Regression Models

Model	AIC	BIC
Poisson	1,203,647.7	1,203,658.1
Negative Binomial (NegBin)	688,925.8	688,946.6
Zero-Inflated Poisson (ZIP)	1,026,842.7	1,026,863.5
Zero-Inflated Negative Binomial (ZINB)	688,927.9	688,959.0
Hurdle Poisson	1,026,842.7	1,026,863.5
Hurdle Negative Binomial	674,058.0	674,089.1

Notes: Lower values of AIC (Akaike Information Criterion) and BIC (Bayesian Information Criterion) indicate better model fit. The Hurdle Negative Binomial model has the lowest AIC and BIC among the candidate models, while the Negative Binomial model has the second lowest AIC and BIC among all candidates.

and month-fixed effects while using population density as an offset. Table 2 supports this specification: the NB specification (AIC $\approx$ 688,926; BIC $\approx$ 688,947) substantially outperformed alternative approaches: it reduced information loss by over 500,000 AIC units compared to the Poisson baseline and exhibited superior fit relative to zero-inflated Poisson (ZIP) and zero-inflated negative binomial (ZINB) models by comparable margins.

Our primary specifications at the census division scale include two distinct formulations. The first, a raw model, regresses unlogged expected nighttime crime counts on average nighttime light (NTL) intensity:

$$\begin{aligned} \text{Expected Crime}_{it} = & \alpha_i + \gamma \text{Average NTL}_{it} \\ & + \beta_1 \text{Average NTL}_{it} \times \text{Downtown}_{it} \\ & + \beta^\top X_{it} + \varepsilon_{it} \end{aligned}$$

where α_i denotes division-specific fixed effects. The second, a log-log model, facilitates elasticity interpretation by expressing variables in natural logarithms:

$$\begin{aligned} \log(\text{Expected Crime}_{it}) = & \alpha_i + \gamma \log(\text{Average NTL}_{it}) \\ & + \beta_1 \log(\text{Average NTL}_{it}) \times \text{Downtown}_{it} \\ & + \beta^\top X_{it} + \varepsilon_{it} \end{aligned}$$

Here, in both model specifications, Expected Crime serves as the outcome variable, and Average NTL represents the treatment variable of interest, for which we also include the interaction term Average NTL $\times$ Downtown to account for the potential heterogeneity effects. In addition, covariates X_{it} , including Year, Month fixed effects, Daytime Expected Crime, population density, and Average

Income (scaled to \$100,000 units), are included to support the CIA.

To address potential heteroskedasticity identified by White tests across both specifications, we report cluster-robust standard errors grouped at the census division level. Additionally, we conducted Ramsey RESET tests to evaluate the linearity assumption. For robustness, we replicated analyses using unadjusted raw crime counts as the outcome variable; results remained consistent, supporting the validity of our approach. By leveraging fine-grained census division data, this analysis enables the detection of micro-level variations and nuanced spatial heterogeneity in luminosity effects.

NEIGHBORHOODS

Given the absence of zero-inflated observations in monthly nighttime crime counts at the neighborhood level, we retained the unadjusted outcome variable and estimated two model specifications. The first, a linear model, regresses raw crime counts on nighttime light (NTL) intensity:

$$\begin{aligned} \text{Crime Count}_{it} = & \alpha_i + \gamma \text{Average NTL}_{it} \\ & + \beta_1 \text{Average NTL}_{it} \times \text{Downtown}_{it} \\ & + \beta^\top X_{it} + \varepsilon_{it} \end{aligned}$$

where α_i denotes neighborhood-specific fixed effects. The second, a log-log specification, facilitates elasticity interpretation:

$$\begin{aligned} \log(\text{Crime Count}_{it}) = & \alpha_i + \gamma \log(\text{Average NTL}_{it}) \\ & + \beta_1 \log(\text{Average NTL}_{it}) \times \text{Downtown}_{it} \\ & + \beta^\top X_{it} + \varepsilon_{it} \end{aligned}$$

Similarly, in both model specifications, Crime Count serves as the outcome variable, Average NTL as the treatment variable, and the interaction term Average NTL $\times$ Downtown captures potential spatial heterogeneity. Covariates X_{it} include year and month fixed effects, daytime crime counts, population density, and regional average income (scaled in \$100,000 units).

Consistent with the census division analysis, White tests revealed heteroskedasticity in both specifications. We addressed this by estimating cluster-robust standard errors, grouped at the neighborhood level. Ramsey RESET tests further confirmed the adequacy of the linear functional form.

The neighborhood scale offers two key analytical benefits. First, the larger spatial scale reduces measurement error in monthly NTL estimates, enhancing the precision of luminosity effect detection. Second, this enables better control of localized spillover effects (e.g., radiation of nighttime activity from downtown cores to adjacent areas).

HETEROGENEITY CHECKS

We conducted two heterogeneity checks: the first analysis considered the potential different effects of luminosity on crimes between the Downtown area and other urban territories. Specifically, we re-estimated our regressions in two different scales, excluding the Average NTL $\times$ Downtown interaction, but on datasets with solely observations in downtown and solely observations in other urban territories. This approach validated our hypothesis that downtown has a different crime pattern and substantiated our decision to include a downtown control in our main analysis.

The second heterogeneity check re-estimated the neighborhood-level regressions across various crime types to determine whether increases in luminosity exert disparate impacts on different types of criminal activity. It not only reinforced the overall findings of our main models but also provided additional insights on future research and policy prescriptions.

In a word, both heterogeneity checks were implemented with the same set of covariates (daytime crime, income, population density), and cluster-robust standard errors as in our primary specifications, ensuring methodological consistency. The stratified results further underscore the importance of spatial and typological analysis in urban crime research.

III. Analysis

A. Data and Assumptions

Our datasets of the census and neighborhood data include time and location-based fixed effects, as well as covariates for population density, income, daytime crime count, and a 'Downtown' indicator. This uniform structure in terms of covariates ensures consistency for comparative analysis. We observe clear heterogeneity in crime patterns across neighborhoods. Downtown stands out as a significant outlier, with both the highest average crime count and the greatest standard deviation, indicating not only more crime but also greater fluctuation over time. Figure 1 illustrates this pattern well, with Downtown clearly standing out among the top five neighborhoods in terms of crime. This supports the inclusion of a Downtown indicator variable in our analysis to account for its distinct

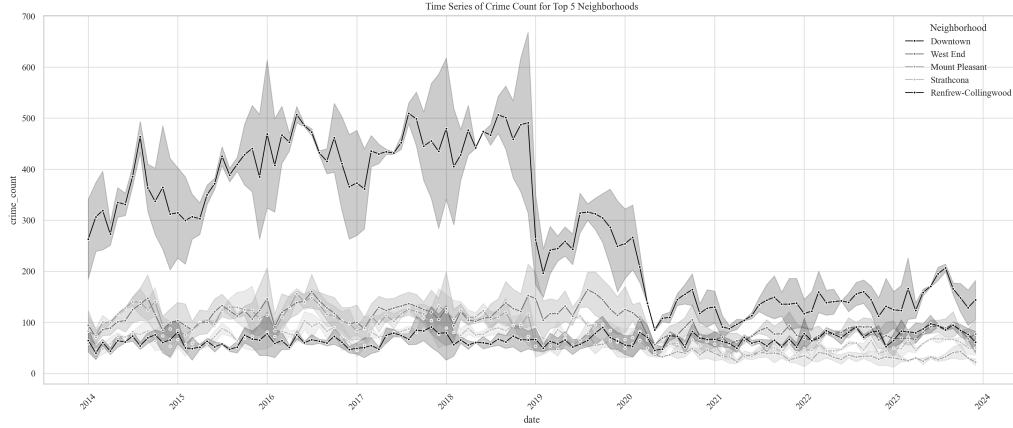


Figure 1. : Time Series of Crime Counts Across Top 5 Vancouver Neighborhoods.

crime dynamics.

Day and night observations originate from the same locations. Supporting this, the data contains identical attributes (e.g., neighborhood, year, month fixed effects, population density, and income) across day and night samples, with no differences. The only exception is crime counts, which are higher at night (mean difference: 20.57, +51.4%). Covariate balance and fairness checks in Table 1 further validate this assumption.

The causal identification in our study rests on the assumption that once we partial out region-based and time-based fixed effects, time-x-region fixed effects, and a small set of theoretically motivated controls, month-to-month variation in VIIRS radiance is as-good-as-random (CIA). In particular, daytime-crime counts proxy unobservable foot-traffic shocks (Domínguez & Asahi 2019; Doleac & Sanders 2015), income and density absorb socioeconomic targets, and a is-downtown term allows for the control of decreasing marginal returns on lighting in the Vancouver central business district area. Reverse causality concerns are mitigated by the assumption that short-term crime fluctuations do not significantly affect light levels.

B. Regression Analysis

Our results at both scales consistently show a significant negative relationship between night-time light intensity and crime. In panel linear regressions at Census Division Scale, higher NTL is associated with lower crime, with coefficients ranging from -0.015^{***} to -0.025^{***} . Log-log models confirm this relationship, showing elasticities from -0.059^{***} to -0.092^{***} . While at the Neighborhood Scale, the linear models have coefficients ranging from the insignificant 0.363 to

(a) Regression of Expected Crime on Average NTL at Census Divisions Scale

	Expected Crime			Log(Expected Crime)		
	(1)	(2)	(3)	(4)	(5)	(6)
Average NTL	-0.015*** (0.004)	-0.021*** (0.004)	-0.025*** (0.003)			
Log(Average NTL)				-0.059*** (0.003)	-0.087*** (0.004)	-0.092*** (0.004)
Expected Day Crime			0.456*** (0.130)			0.052*** (0.017)
Population Density		0.0001*** (0.00002)	0.0001*** (0.00002)	0.00005*** (0.00001)	0.00004*** (0.00001)	0.00004*** (0.00001)
Income (in \$100,000)		-1.165*** (0.113)	-0.788*** (0.126)	-0.025 (0.054)	-1.183*** (0.034)	-1.139*** (0.036)
Average NTL $\times$ Downtown	0.012 (0.009)	0.019** (0.009)	0.031*** (0.006)			
Log(Average NTL) $\times$ Downtown				0.062*** (0.009)	0.096*** (0.012)	0.105*** (0.012)
Year Fixed Effects	Yes	No	No	Yes	No	No
Month Fixed Effects	Yes	No	No	Yes	No	No
Observations	107,249	107,249	107,249	107,249	107,249	107,249
R ²	0.072	0.055	0.333	0.538	0.327	0.354
Adjusted R ²	0.063	0.046	0.326	0.534	0.320	0.348

Notes: Robust standard errors clustered by census division are reported in parentheses. Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Models (1)–(3) use the expected number of crimes per division; Models (4)–(6) use its logarithm. “Average NTL” is nighttime-light intensity. “Downtown” is a binary indicator for divisions in the urban core. Year and month fixed effects are included in Models (1) and (4) only but omitted from the table.

(b) Regression of Crime Count on Average NTL at Neighborhood Scale

	Crime Count			Log(Crime Count)		
	(1)	(2)	(3)	(4)	(5)	(6)
Average NTL	0.363 (1.604)	-1.318 (0.994)	-3.463*** (0.521)			
Log(Average NTL)				-0.049 (0.042)	-0.162*** (0.039)	-0.211*** (0.038)
Daytime Crime Count			0.942*** (0.023)			0.004*** (0.001)
Population Density	-0.075* (0.045)	-0.078 (0.049)	-0.018 (0.013)	-0.0003 (0.0002)	-0.0004** (0.0002)	-0.0001 (0.0001)
Income (in \$100,000)	108.239 (230.162)	-7.335 (37.596)	-8.562 (12.443)	0.666 (1.693)	-1.137*** (0.289)	-1.142*** (0.268)
Average NTL $\times$ Downtown	-0.468 (1.497)	1.310 (0.887)	4.274*** (0.497)			
Log(Average NTL) $\times$ Downtown				0.081* (0.045)	0.210*** (0.039)	0.261*** (0.039)
Year Fixed Effects	Yes	No	No	Yes	No	No
Month Fixed Effects	Yes	No	No	Yes	No	No
Observations	2,420	2,420	2,420	2,420	2,420	2,420
R ²	0.302	0.248	0.696	0.350	0.195	0.286
Adjusted R ²	0.289	0.241	0.692	0.338	0.187	0.278

Notes: Robust standard errors clustered by group are reported in parentheses. Significance levels are denoted by * $p < 0.1$, ** $p < 0.05$, and *** $p < 0.01$. The dependent variable in Models (1)–(3) is the number of crimes per unit, while the dependent variable in Models (4)–(6) is the log number of crimes per unit. Average NTL refers to nighttime light intensity. “Downtown” is an indicator for downtown locations. Year and month fixed effects are included in Models (1) and (4) only.

Table 3—: Regressions of Crime on Average NTL at Two Scales

-3.463***, with log-log elasticity ranging from -0.049 to -0.211***. However, the downtown interaction is positive and significant in log-log models in both scales, suggesting weaker effects in dense urban areas. Overall, the findings are robust across specifications and support the conclusion that increased lighting is linked to reduced crime.

To address reverse causality and endogeneity, we use an expected crime model as our independent variable. The results remain robust across specifications. Average NTL is consistently negative and significant, even after controlling for population density, income, and fixed effects. A log-log framework confirms this. The coefficient for $\log(\text{Average NTL})$ is negative and significant (-0.092*** in column 3), while the downtown interaction term is positive, indicating spatial heterogeneity. The adjusted R^2 rises to 0.348 with controls, reinforcing explanatory power.

Our analysis reveals robust and heterogeneous effects of nighttime lights (NTL) on crime when distinguishing between downtown and urban areas. In downtown cores, NTL shows a positive or neutral association with crime (e.g., +1.024*** for crime counts), likely reflecting denser activity and reporting. In contrast, urban territories exhibit strong negative relationships (-0.098*** for log crime), consistent with broader deterrent effects of lighting. Controls like daytime crime and income remain significant, with higher population density correlating positively with crime in urban areas but negatively downtown. These results underscore spatial nuance: while NTL may signal crime hotspots downtown, it inversely predicts crime elsewhere. The findings hold across geographic scales (neighborhoods/census divisions) and specifications (levels/logs), supporting targeted policy design. In dense downtown areas, excessive existing light could diminish the marginal deterrent effect of additional illumination, while in less saturated urban areas, increased lighting remains negatively associated with crime. This suggests an optimal threshold where lighting effectively deters crime, beyond which benefits plateau due to factors like high population density and activity levels.

C. Heterogeneity Analysis

In addition, we further analyze heterogeneity across crime types. reveals that offenses like "Other Theft" and "Theft from Vehicle" are more frequent and variable in downtown areas, while crimes such as "Break and Enter Residential/Other" and "Theft of Vehicle" remain low and stable across regions. These differences suggest that crime types may vary in their relationship to lighting.

We further explore how crime types vary in their relationship with average NTL. The scatterplots (Figure B4) show distinct patterns: crimes like "Other Theft," "Break and Enter Commercial," and "Theft from Vehicle" exhibit strong positive

Table 4—: Regression of Expected Crime on Average NTL by Downtown vs. Urban Territories at Census Divisions Level

	Expected Crime		Log(Expected Crime)	
	Downtown	Urban Territory	Downtown	Urban Territory
Average NTL	0.004 (0.005)	-0.027*** (0.003)		
Log(Average NTL)			0.005 (0.010)	-0.098*** (0.004)
Expected Day Crime	0.421** (0.187)	0.514*** (0.093)	0.025* (0.013)	0.102*** (0.031)
Population Density	0.0001* (0.00003)	0.0001*** (0.00001)	0.00002*** (0.00000)	0.0001*** (0.00001)
Income (in \$100,000)	-3.097* (1.810)	-0.677*** (0.078)	-1.037*** (0.187)	-1.117*** (0.036)
Observations	5,444	101,585	5,444	101,585
R ²	0.284	0.428	0.274	0.388
Adjusted R ²	0.276	0.423	0.267	0.382

Notes: Robust standard errors clustered by group are reported in parentheses. Significance levels are denoted by * $p < 0.1$, ** $p < 0.05$, and *** $p < 0.01$. The dependent variable is the expected number of crimes per unit in columns (1)–(2), and log-transformed in columns (3)–(4). This table presents a robustness check at the census division level, showing heterogeneous effects for downtown versus urban areas. Data sources: VPD Open Data and NASA’s Black Marble.

correlations with the average NTL, while "Break and Enter Residential/Other" and "Theft of Vehicle" show weak or negligible associations. These differences highlight how certain crimes are more sensitive to lighting conditions than others.

To take into account this heterogeneity, we run a model conditioning on each crime type. Some crimes have a differential relationship to light, which should be taken into consideration for robustness. Our findings demonstrate consistent effects of average NTL across diverse crime categories. As shown, commercial burglary (-0.278***), residential burglary (-0.188***), and theft from vehicle (-0.223***) exhibit particularly strong negative associations with illumination, mirroring our core results. Notably, bicycle theft shows no significant relationship (-0.089), suggesting lighting’s deterrent effects are crime-specific. The downtown interaction terms further confirm spatial heterogeneity, with residential burglary displaying the strongest positive moderation (0.539***) - indicating lighting’s reduced effectiveness in dense urban cores. These patterns persist even when controlling for daytime crime counts, reinforcing that our main results are not artifacts of general crime trends but reflect lighting’s distinct nocturnal impacts.

Table 5—: Regression of Crime Count on Average NTL by Downtown vs. Urban Territories at Neighborhood Level

	Crime Count		Log(Crime Count)	
	Downtown	Urban Territory	Downtown	Urban Territory
Average NTL	1.024*** (0.000)	-3.491*** (0.410)		
Log(Average NTL)			0.052*** (0.000)	-0.266*** (0.040)
Daytime Crime Count	0.554	0.812*** (0.061)	0.002*** (0.000)	0.012*** (0.001)
Population Density	-0.162	-0.007 (0.005)	-0.0005*** (0.000)	0.00000 (0.0001)
Income (in \$100,000)		-19.596** (8.632)		-0.929*** (0.232)
Observations	110	2,310	110	2,310
R ²	0.821	0.485	0.804	0.306
Adjusted R ²	0.816	0.479	0.798	0.298

Notes: Robust standard errors clustered by group are reported in parentheses. Significance levels are denoted by * $p < 0.1$, ** $p < 0.05$, and *** $p < 0.01$. The dependent variable is the number of crimes per unit in columns (1)–(2), and log-transformed in columns (3)–(4). This table presents a robustness check at the neighborhood level, showing heterogeneous effects for downtown versus urban territories. Data sources: VPD Open Data and NASA’s Black Marble.

Table 6—: Log-Log Heterogeneous Regression of Different Crime Types on Average NTL in the Scale of Neighborhoods

	Break and Enter Comm.	Break and Enter Resid./Other	Other Theft	Theft of Vehicle	Theft of Bicycle	Theft from Vehicle
Log(Average NTL)	-0.278*** (0.063)	-0.188*** (0.050)	-0.176*** (0.042)	-0.117 (0.073)	-0.089 (0.082)	-0.223*** (0.039)
Daytime Crime Count	0.233*** (0.021)	0.268*** (0.023)	0.302*** (0.050)	0.211*** (0.032)	0.555*** (0.036)	0.417*** (0.032)
Population Density	0.00000 (0.00004)	0.0001 (0.0001)	-0.0001 (0.0001)	0.00001 (0.0001)	-0.0001 (0.0001)	-0.0001** (0.00004)
Income (in \$100,000)	-0.096 (0.238)	-1.098*** (0.237)	0.548* (0.308)	-1.177*** (0.162)	-0.153 (0.154)	-0.772*** (0.184)
Log(Average NTL) $\times$ Downtown	0.149** (0.064)	0.539*** (0.050)	0.177*** (0.041)	-0.030 (0.073)	-0.021 (0.082)	0.302*** (0.038)
Observations	1,385	2,082	2,308	1,310	1,354	2,311
R ²	0.055	0.147	0.099	0.093	0.337	0.251
Adjusted R ²	0.037	0.136	0.088	0.075	0.324	0.243

Notes: Robust standard errors clustered by group are reported in parentheses. Significance levels are denoted by * $p < 0.1$, ** $p < 0.05$, and *** $p < 0.01$. This table presents a heterogeneity analysis of log-transformed crime counts across crime types at the neighborhood level. Independent variables include log(Average NTL), interaction with Downtown dummy, and control variables. Data sources: VPD Open Data and NASA’s Black Marble.

The findings suggest that increased nighttime illumination is associated with reduced crime rates in urban areas, consistent with existing literature on the deterrent effect of lighting. However, the diminishing returns observed in densely lit downtown cores highlight the importance of context-specific lighting policies. Although our study advances the literature by researching a robust link between nighttime luminosity and crime in Vancouver, several limitations warrant discussion and point the way for future research. Following are some recognition of how it can be further improved and how subsequent work could address these gaps.

As Table 2 shows, a hurdle negative binomial marginally outperforms our preferred negative binomial model. We rejected the hurdle specification because its extra zero-inflation equation has no clear behavioral analogue in our setting. However, the performance gap suggests that a two-process model might still capture meaningful variation. Future work should formally test for excess zeros.

Next, our covariate set is largely static, relying on population density, a downtown indicator, and daytime crime to proxy target availability. It omits real-time measure of foot traffic and vehicle inflows that are highly relevant to vehicle break-ins that both influence and are influenced by ambient light. Adding data sources such as traffic-congestion indices and transit-boarding counts would help future studies isolate the effect of luminosity from fluctuations in the number of people, vehicles, or properties exposed to victimization. Furthermore, we control for neighbourhood characteristics with cross-sectional census snapshots (2016 and 2021). While justifiable given census frequency, this practice overlooks rapid local changes such as condo boom or pandemic residential shifts that might independently alter both luminosity and crime rate. To address this, future studies could use proxies of socioeconomic dynamics that occur within census cycles.

IV. Discussion and Extensions

We observed significant results at different scales. With full controls, at the census divisions scale, the coefficient of the linear model is -0.025^{***} , while the log-log model has a coefficient of -0.092^{***} . At the scale of neighborhoods, the coefficient of the linear model is -3.463^{***} , while the log-log model has a coefficient -0.211^{***} . As for the heterogeneity analysis omitting Downtown observations, at two different scales, we observed coefficients of -0.027^{***} and -3.491^{***} for linear models, and -0.098^{***} and -0.266^{***} for log-log models. The heterogeneity analysis on different types of crimes also all showed negative results, we have Break and Enter Commercial (-0.278^{***}), Break and Enter Residential/Other (-0.188^{***}), Other Theft (-0.176^{***}), Theft of Vehicle (-0.117), Theft of Bicycle (-0.089) and Theft from Vehicle (-0.223^{***}).

However, we note several issues in our analysis that may potentially induce heterogeneity. First, the parametric form of our regression model implicitly as-

Table 7—: RESET Test on the Scale of Census Divisions

Variable	Expected Crime (1)	Log(Expected Crime) (2)
Average NTL	−0.004 (0.004)	
Log(Average NTL)		−0.016 (0.010)
Expected Day Crime	0.190 (0.131)	0.048*** (0.014)
Population Density	−0.00000 (0.00002)	0.00001** (0.00001)
Income (in \$100,000)	0.087 (0.177)	−0.094 (0.139)
Fitted Value $\hat{y}$	0.951*** (0.186)	0.952*** (0.118)
Fitted Value Squared $\hat{y}^2$	0.004 (0.027)	−0.374 (0.228)
Fitted Value Cubed $\hat{y}^3$	−0.011*** (0.003)	−1.945*** (0.449)
Observations	107,249	107,249
R ²	0.382	0.384
Adjusted R ²	0.376	0.378

Notes: Standard errors clustered by census division appear in parentheses. Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. This RESET test augments each baseline regression with powers of the fitted value ($\hat{y}$, $\hat{y}^2$, $\hat{y}^3$) to detect functional-form misspecification. Model (1) estimates levels of expected crime; Model (2) estimates its logarithm. Year and month fixed effects are included in both models but omitted here for brevity.

sumes that the CEF itself is linear. However, Table 5 shows that returns on light decrease drastically at the highest levels of luminosity (downtown), showing a divergence in marginal effects. The log-log model hence incorrectly assumes that β_1 remains constant regardless of luminosity levels. This is further verified by an additional Ramsey RESET test (Table 7), which shows that we rejected the null hypothesis that the CEF being linear and correctly specified. A better regression specification would have been a non-linear model such as a piecewise linear regression with breaks at higher levels of luminosity. Nevertheless, based on the Information-Criterion Model Selection Test (Table 8) the Bayesian Information Criterion (BIC) still favors the simple log-log linear model—its low parameter count makes it the most parsimonious specification and likely the most robust for out-of-sample prediction. At the same time, the AIC-optimal “Spline df3” natural spline (with three degrees of freedom) offers an attractive middle ground: it captures the moderate non-linear curvature in the luminosity–crime relationship without the instability of higher-order polynomials, making it a promising candidate when genuine non-linear effects are anticipated.

Furthermore, although the NASA Black Marble dataset already applies bidi-

Table 8—: Specification Comparisons

Model	n	k	SSR	AIC	BIC
Linear	2420	5	234.7337	-5636.032	-5607.075
Quad	2420	6	234.6547	-5634.847	-5600.097
Cubic	2420	7	234.5711	-5633.709	-5593.168
Spline df3	2420	9	233.5344	-5640.428	-5588.304
Spline df4	2420	10	233.5448	-5638.320	-5580.405
AIC Best Model: Spline df3					
BIC Best Model: Linear					

rectional reflectance distribution function (BRDF) corrections for snow cover and lunar-phase effects, residual biases can still remain. In our current project we relied on the **blackmarbler** R package, which downloads VNP46A2 files from the Archive 5,000 collection, a first-generation, land-only, 16-bit product. Future work could instead draw on Archive 5,200, the re-processed float-32 collection released in late 2024 that offers cross-sensor calibration and other improvements.

Additionally, the central assumption to our paper is the conditional independence assumption. We argue that, conditional on the set of fixed effects and socioeconomic covariates, luminosity values are as good as random with respect to nighttime crime potential outcomes. One of the key covariates required for the CIA to hold is economic activity: economic activity both provides financial incentives for criminals to commit crime, and the development spurred by economic activity increases luminosity values. However, the city of Vancouver does not provide detailed data on economic activity within each neighborhood, requiring us to control for economic activity some other way. Hence, we assume two things: first, that changes in movement patterns cause changes in economic activity, a well-researched topic. Second, we assume that changes in daytime crime cause changes in movement patterns. Felson & Boivin (2015) and Paramasivan et al. (2024) provide some evidence for this. Hence, by controlling for daytime crime, we control for movement patterns, and consequently economic activity. Key to this assumption is that the causal diagram between daytime crime, movement patterns, and economic activity is unidirectional. However, research indicates that this is not likely to be true: During resource booms and other positive economic shocks, cities often experience increased crime (Soares & Souza, 2025; Gourley & Madonia, 2018). In fact, more evidence supports the notion that the causal chains between daytime crime, movement patterns, and economic activity is bidirectional. If this is the case, then not only would including daytime crime as a covariate not help control for economic activity, but, as daytime crime is a descendant, controlling for daytime crime results in Berkson’s paradox.

Lastly, our paper assumes that short-term crime fluctuations do not significantly affect light levels, hence mitigating any potential reverse causality concerns. However, this assumption may not necessarily be true. For instance, businesses may close early or turn signage off after surges in robberies or street crime. Likewise, police departments in the united states regularly install flood-lights in areas with high spikes in crime. For instance, permanent flood light towers had been installed in NYC housing developments well before the Chalfin et al. (2022) study (Chiel, 2017) however, whether or not such policies are implemented in Vancouver specifically remains undisclosed.

A. Extensions

While analysis of crime using satellite data remains promising, future research should focus on non-linear models and on cities which provide extensive economic and population traffic data to address potential endogeneity issues. Namely,

- 1) Future studies should include specific covariates for both economic activity and daily population mobility. Specifically, since crime in-and-of-itself also causes changes in economic activity and population mobility, researches should be careful in how they include these controls. For example, researchers could use one-to-two month pre-crime lags as a control instead of current economic and mobility data, preventing crime from becoming a potential collider.
- 2) Additionally, researchers would need to fit a non-linear model on the data to account for different marginal returns on light between downtown and non-downtown areas. For instance, a thin-plate spline regression could be fitted onto our luminosity variable, creating a flexible, smooth function and allowing for $\partial^2(f)/\partial(AverageNTL_{it})^2$ to be any smooth function rather than having a fixed power-law curvature. (Columbia Public Health, 2023; Eberly, 2023).
- 3) Furthermore, researchers would need to find a way to overcome potential reverse causality between nighttime light and crime. The most straightforward solution would be to implement a IV-based approach, using data on lunar luminance, which would affect the luminosity captured in satellite data but not crime rates. Researches could then try running a panel 2SLS setup by first regressing luminosity on lunar luminance and then using the fitted values to estimate crime effects.

Hence, a full model that would account for these heterogeneity issues would look something like:

$$\begin{aligned}
\text{NTL}_{it} &= \alpha_i + \pi_1 \text{LunarLum}_{it} \\
&+ \pi_2 \text{Econ}_{i,t-1} \\
&+ \pi_4 \text{Mobility}_{i,t-1} + u_{it}
\end{aligned}$$

Then, we estimate Crime as:

$$\begin{aligned}
\text{Crime}_{it} &= \sigma_i + f(\widehat{\text{NTL}}_{it}) \\
&+ \beta_1 \text{Econ}_{i,t-1} \\
&+ \beta_2 \text{Mobility}_{i,t-1} + \varepsilon_{it}
\end{aligned}$$

Where $f(\cdot)$ is a non-linear function.

V. Conclusion

Our study provides robust evidence that increased nighttime illumination is associated with reduced crime rates in urban areas, aligning with existing literature on the deterrent effects of lighting. However, the relationship exhibits significant spatial heterogeneity: while brighter lighting correlates with lower crime in most urban areas, densely lit downtown cores show diminishing returns, likely due to saturation effects and unique environmental factors like high population density, foot traffic and activity levels.

In areas with moderate baseline lighting, strategic increases can enhance public safety cost-effectively. In contrast, downtown areas may require alternative crime prevention measures, as additional lighting yields limited marginal benefits. Future research should explore optimal illumination thresholds and complementary interventions to maximize crime reduction across diverse urban environments. Future research should further quantify the non-linear characteristics of the lighting-crime relationship by testing alternative functional forms such as spline or quadratic models across diverse urban contexts and also consider various fully saturated covariates. Additionally, comparative studies across different cities are needed to examine the generalizability of these patterns beyond Vancouver. Further, research should investigate the localized (Vancouver) effects of daylight saving time (DST) transitions on crime rates, building upon prior DST-crime literature. Furthermore, the effect of mobility needs to be considered in further research.

For policymakers, our results support lighting as an effective safety intervention in moderately lit areas, while highlighting the need for complementary strategies

in saturated downtown cores. Implementation should be guided by localized cost-benefit analyses that account for these non-linear effects.

Generative-AI Disclosure

We used ChatGPT as a research assistant to locate additional peer reviewed papers (ChatGPT's "deep research" function), after a comprehensive manual search, on additional background literature, as well as literature related to preprocessing sparse crime count data using Poisson models and negative binomial variants. Furthermore, it was used to check how our chosen covariates, particularly cross-sectional census indicators, have been used in other panel fixed-effects studies, and to polish grammar and wording. The tool also suggested alternative ways to visualise the basic distributional characteristics of our covariates and outcome variables, as well as additional tests that could be run to verify the suspected non-linearity of our regression, and helped us understand the math behind some potential contenders for non-linear regression models.

Every AI suggestion was extensively reviewed prior to inclusion by every author. ChatGPT is not listed as an author and bears no responsibility for the work, and the authors accept full accountability for the integrity of the content of the paper, in line with SAGE's Artificial Intelligence Policy.

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APPENDIX A: STATISTICS AND ADDITIONAL TABLES

Table A1—: Summary Statistics of Census Divisions Covariates

Variable	Mean	Std. Dev.	Min	Max
Electric Index	115.10	7.23	94.80	122.50
Income (scaled by \$100k)	0.81	0.26	0.13	2.22
Expected Day Crime	0.91	3.34	0.03	128.44
Population Density	10 102.63	9 822.78	281.80	77 692.30
Immigrant Proportion	0.42	0.14	0.05	0.88
University Proportion	0.18	0.18	0.00	0.66
School Index	2.62	0.81	0.00	7.91
Bus Index	2.91	0.78	0.29	6.66
Cultural Space Index	2.21	0.94	0.00	7.48
Mean Land Value	1.96	3.84	0.06	64.83
Downtown (count = 107,249)				
Yes		5,444		
No		101,805		

Notes: Sample size $N = 107,249$. Continuous covariates are summarized by mean, standard deviation, minimum, and maximum. Panel B reports the counts for the Downtown indicator.

Table A2—: Summary Statistics of Neighborhood Covariates

Variable	Mean	Std. Dev.	Min	Max
Income (scaled by \$100k)	0.8468	0.2089	0.3387	1.4092
Day Crime Count	39.4901	57.7201	0	523
Population Density	6 459.987	4 931.169	1 853.673	23 663.51
Downtown (count = 2,420)				
Yes		110		
No		2,310		

Notes: Sample size $N = 2,420$. Continuous covariates are summarized by mean, standard deviation, minimum, and maximum. Panel B reports the counts for the Downtown indicator.

Table A3—: Summary Statistics of Crime Counts by Neighborhood

Neighborhood	Obs.	Mean	Median	Min	Max	Std. Dev.
Downtown	240	289.80	274.0	69	668	149.25
West End	240	102.87	97.5	33	214	37.47
Mount Pleasant	240	75.94	71.0	12	171	42.48
Strathcona	240	67.25	65.0	27	130	22.61
Renfrew–Collingwood	240	66.23	65.0	26	133	20.30
Grandview–Woodland	240	60.29	57.0	15	136	23.44
Fairview	240	58.45	57.0	15	134	22.41
Kitsilano	240	55.52	51.5	15	122	23.62
Kensington–Cedar Cottage	240	49.54	48.0	13	98	18.23
Hastings–Sunrise	240	38.41	34.0	7	90	18.53
Sunset	240	34.53	31.0	8	87	14.58
Marpole	240	31.20	29.0	11	63	11.50
Riley Park	240	30.36	29.0	5	74	14.67
Killarney	240	23.60	22.0	7	64	8.87
Victoria–Fraserview	240	20.34	20.0	4	45	8.36
Oakridge	240	18.48	18.0	3	47	8.47
Dunbar–Southlands	240	16.29	15.0	2	58	9.16
South Cambie	240	14.70	14.0	1	36	6.46
Kerrisdale	240	13.83	13.0	1	42	6.30
West Point Grey	240	13.52	12.5	1	42	7.71
Shaughnessy	240	12.92	12.0	2	40	7.20
Arbutus Ridge	240	12.10	11.0	0	31	5.83

Notes: Monthly crime counts (2010–2029) across 22 Vancouver neighborhoods (240 observations each). Reports mean, median, min, max, and standard deviation.

Table A4—: Robustness Check: Regression of Crime Count on Average NTL at the Census Division Level

	(1)	(2)	(3)
Average NTL	-0.001 (0.007)	-0.007 (0.007)	-0.017*** (0.004)
Daytime Crime Count			0.406*** (0.003)
Population Density	0.00001 (0.00001)	0.00000 (0.00001)	0.00001 (0.00000)
Income (Scaled by \$100,000)	-0.009 (0.154)	-1.025*** (0.153)	-0.751*** (0.053)
Average NTL $\times$ Downtown	0.024 (0.016)	0.030* (0.016)	0.048*** (0.005)
Year Fixed Effects	Yes	No	No
Month Fixed Effects	Yes	No	No
Observations		107,249	
R ²	0.006	0.004	0.130
Adjusted R ²	-0.003	-0.006	0.122

Notes: Robust standard errors clustered by census division in parentheses. Significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Model (1) includes year and month fixed effects; omitted for brevity in Models (2)–(3).

APPENDIX B: VISUALIZATIONS

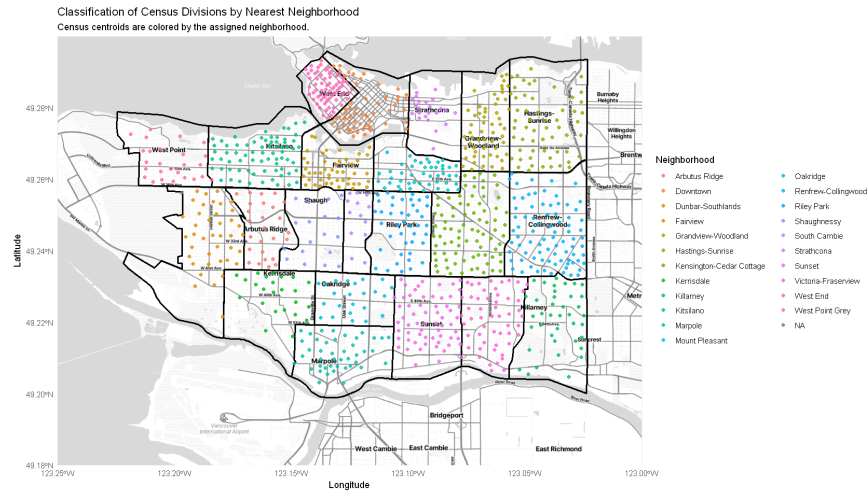


Figure B1. : Classification of Census Divisions into Neighborhoods by Geospatial Centroids

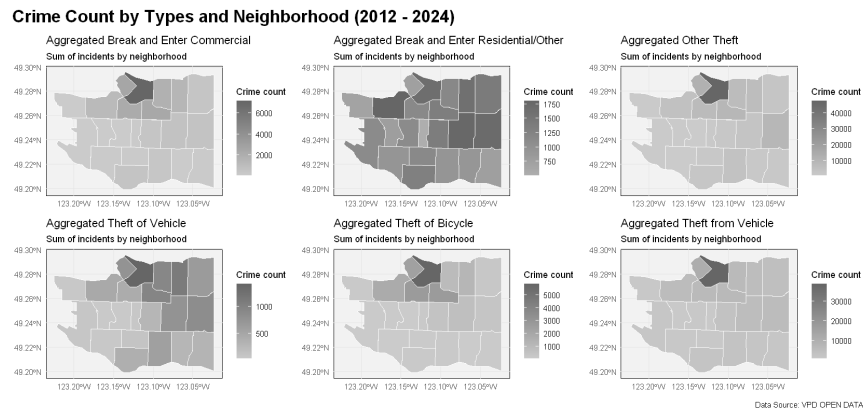


Figure B2. : Crime Counts by Types and Neighborhoods Across 2014–2023

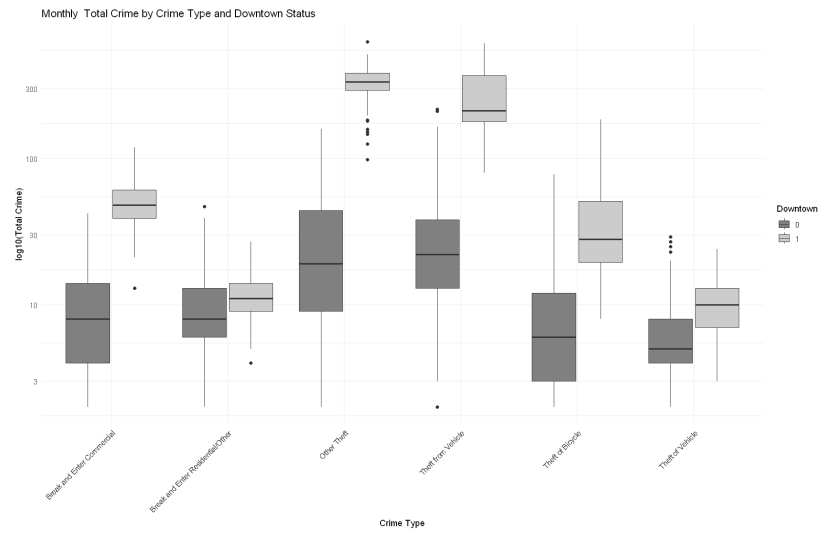


Figure B3. : The Total Crime by Type and Downtown Across 2014–2023

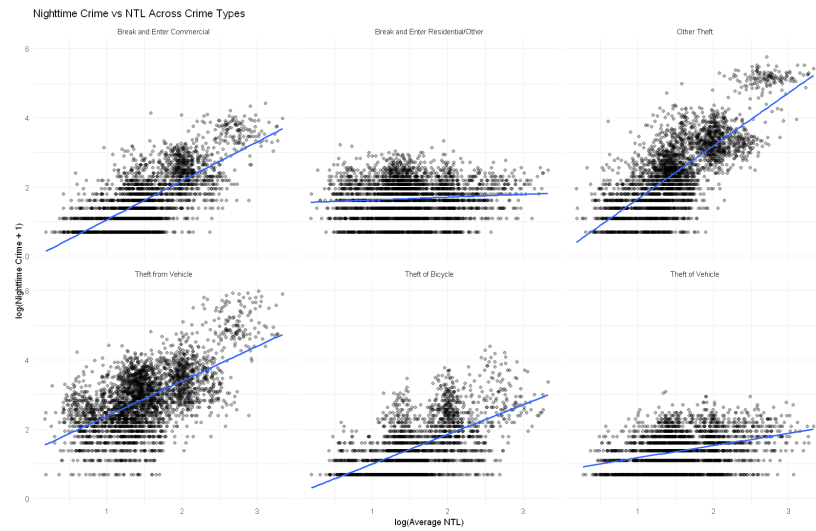


Figure B4. : The Distribution of Crime Counts Against Average NTL by Types